

RF HiRes Dash — Technical Reference

Propagation Physics, Formulas & Calculation Methods

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1. ITU-R P.368-9 Propagation Model

ITU-R Recommendation P.368-9 is an international technical standard for predicting ground wave propagation for frequencies from 0.15 MHz to 30 MHz (LF, MF, and HF bands).

What is ITU-R P.368-9?

"Ground wave" means the radio signal travels close to the Earth's surface, following its curvature. This dominates at short-to-medium distances (<1,000 km) and low-to-medium frequencies, making it ideal for broadcasting, maritime, and amateur radio.

Valid Frequency Range

0.15 MHz to 30 MHz

- 0.15–0.535 MHz: LF (low frequency)
- 0.535–1.7 MHz: MF (medium frequency, AM broadcast)
- 1.7–30 MHz: HF (high frequency, amateur, maritime, aviation)

For frequencies outside this range, use alternative models (e.g., UHF/VHF line-of-sight models for 30+ MHz).

Key Inputs to P.368-9

- Frequency (MHz): 0.15–30
- Transmitter height (m): antenna height above ground
- Receiver height (m): antenna height above ground
- Radiated power (W): effective radiated power
- Ground conductivity (S/m): electrical conductivity at TX and RX
- Distance (km): TX to RX distance
- Elevation profile: terrain heights along the path

Output

Received field strength: E (dBμV/m) — decibels above 1 microvolt per meter. This is the standard metric for RF coverage.

2. Path Loss Fundamentals

Free Space Path Loss

In an ideal environment with no obstacles, path loss follows:

$$L_{fs}(\text{dB}) = 20 \log_{10}(d) + 20 \log_{10}(f) + 32.45$$

where: d = distance in km, f = frequency in MHz

Example: 10 MHz, 100 km in free space

$$L_{fs} = 20 \log_{10}(100) + 20 \log_{10}(10) + 32.45 = 40 + 20 + 32.45 = 92.45 \text{ dB}$$

Practical Path Loss (Ground Wave)

Real-world path loss includes attenuation due to ground interaction:

$$L_{\text{path}}(\text{dB}) = L_{\text{fs}} + A(d, f, \sigma, \epsilon_r, h_{\text{tx}}, h_{\text{rx}})$$

where: A = additional attenuation from ground losses, diffraction, and antenna effects (typically 5–50 dB depending on terrain and conductivity)

P.368-9 computes A numerically using complex wave number and Sommerfeld integrals. It's not a simple formula; instead, it's an iterative calculation.

3. Medium Frequency (MF/LF) Propagation (0.15–30 MHz)

Frequency Bands

- 0.15–0.535 MHz: LF (Low Frequency)
- 0.535–1.6 MHz: MF (Medium Frequency, AM broadcast)
- 1.6–30 MHz: HF (High Frequency, amateur, maritime, aviation)

The Ground Wave Mechanism

At frequencies below 30 MHz, radio signals propagate predominantly as ground waves—energy that travels close to the Earth's surface, following the curvature of the Earth. This occurs because of:

- Refraction in the lower atmosphere (bending the signal downward)
- Electrical interaction with the Earth's surface (which supports wave propagation)
- Signal energy that can reach beyond line-of-sight

The ground wave dominates for short-to-medium distances (<1,000 km) and remains constant day and night, unlike skywave (ionospheric bounce), which varies with atmospheric conditions.

Ground Conductivity (σ) — The Critical Parameter

Electrical conductivity measures how easily current flows through a material. Higher σ = better propagation (lower path loss).

Typical conductivity values:

- Sea water: 5.0 S/m
- Wet soil: 0.01–0.1 S/m
- Average land: 0.005 S/m
- Dry sand: 0.0001 S/m

Ground conductivity is the MOST IMPORTANT parameter affecting path loss (other than distance and frequency). A $10\times$ change in σ can change range by $\pm 30\%$.

Why Separate Conductivity at TX and RX?

The model allows separate conductivity values at TX and RX locations because:

- TX over water ($\sigma = 5 \text{ S/m}$), RX over land ($\sigma = 0.005 \text{ S/m}$): dramatically different propagation
- TX in urban area (higher moisture), RX rural: conductivity varies
- Mixed paths: P.368-9 averages conductivity along the path using Millington integration

ITU-R P.368-9 Millington Calculation Method

The P.368-9 model is NOT a simple formula. Instead, it:

- Uses Sommerfeld integral theory to model wave propagation over lossy ground
- Integrates the complex wave number accounting for ground conductivity
- Applies Millington mixed-path integration for paths with varying ground types
- Includes antenna correction factors (height and frequency dependent)
- Computes diffraction over terrain obstacles from the elevation profile

For RF HiRes Dash:

- Input: Frequency, TX/RX heights, antenna ERP, ground conductivity (σ_{tx} , σ_{rx}), elevation profile (~200 points from SRTM)
- Calculation: Numerical integration of P.368-9 equations
- Output: Field strength E (dB μ V/m) at each RX location
- Computational complexity: $O(N)$ where $N \approx 200$ terrain profile points

Terrain Diffraction in the MF Band

Knife-edge diffraction loss is calculated from the elevation profile using Fresnel zone clearance at the dominant obstacle:

Diffraction Loss (dB) = Function of Fresnel zone clearance at dominant obstacle

At MF frequencies, diffraction is less pronounced than at VHF/UHF but still significant for large terrain obstacles (hills >100 m relief).

4. VHF/UHF Propagation (30 MHz – 2 GHz)

Frequency Range & Applications

30–300 MHz: VHF (Very High Frequency, FM broadcast, mobile)

300 MHz–3 GHz: UHF (Ultra High Frequency, TV, cellular, WiFi)

RF HiRes Dash is validated to 2 GHz; use above 2 GHz experimentally only.

Why Ground Wave Fails at VHF/UHF

At VHF/UHF frequencies, ground wave effects vanish because:

- Wavelength becomes very short (10 m at 30 MHz; 15 cm at 2 GHz)
- Ground conductivity effects negligible relative to free-space loss
- Direct and reflected rays dominate over guided ground wave

As a result, P.368-9 is NOT used above 30 MHz. Instead, the model uses Fresnel-Kirchhoff diffraction + two-ray theory.

Component 1: Fresnel-Kirchhoff Diffraction (Deygout Method)

This models how terrain obstacles block radio waves.

Process:

1. Identify the dominant obstacle from the elevation profile
2. Calculate Fresnel parameter: $v = \sqrt{2} \times (2d_1d_2)/(\lambda D)$ where d_1, d_2 = obstacle distances to TX/RX, D = total distance, λ = wavelength
3. Apply Fresnel-Kirchhoff loss: Loss (dB) = $\max(0, 10 \log_{10}(0.5 + 0.62v))$
4. For multiple obstacles, apply successive diffraction (Bullington method)

Result: Diffraction loss typically 5–50 dB depending on terrain blocking. Frequency dependence: Higher frequencies → shorter wavelengths → larger diffraction loss. Doubling frequency increases diffraction loss by ~6 dB.

Component 2: Two-Ray Ground Reflection

Models constructive and destructive interference between:

- Direct ray: straight path from TX to RX
- Reflected ray: bounces off ground between TX and RX

Phase difference between the two creates constructive interference (enhancement) or destructive interference (fading nulls). This explains the typical VHF/UHF fading pattern observed in mobile coverage.

Component 3: Radio Horizon & Horizon Diffraction

Definition: The maximum distance at which TX and RX are mutually visible (line-of-sight), derived from effective Earth radius $R_e = (4/3) \times 6,371 \text{ km} \approx 8,495 \text{ km}$ (ITU-R P.526-15).

Radio horizon distance (approximate): $d_{\text{horizon}} \text{ (km)} \approx 4.12 \times \sqrt{(h_{\text{tx}} \text{ [m]} + h_{\text{rx}} \text{ [m]})}$

Beyond-horizon paths: If RX is beyond the radio horizon, the signal must diffract over Earth's bulge. This is modeled as diffraction at the horizon point, typically producing 20–60 dB loss depending on frequency and height difference.

Component 4: VHF Thickness Penalty

At VHF/UHF, shorter wavelengths are more sensitive to terrain roughness and obstacles. This is captured via a frequency-dependent penalty:

$$\text{Thickness penalty (dB)} = 0.4 \times (f [\text{MHz}] / 30) \times \sqrt{(\text{diffraction loss [dB]})}$$

Effect: Frequency doubling (e.g., 100 MHz → 200 MHz) increases loss by ~3 dB, capturing reality that UHF suffers more from terrain blocking than VHF.

Combined Path Loss Calculation

For LOS paths (mutual visibility):

- Path Loss (dB) = Free Space Loss + Two-Ray Factor + Diffraction (if partial)

For non-LOS paths (terrain blocking):

- Path Loss (dB) = Free Space Loss + Deygout Diffraction + Horizon Loss (if needed) + VHF Thickness Penalty

Free Space Loss (reference):

- $L_{fs} \text{ (dB)} = 20 \log_{10}(d [\text{km}]) + 20 \log_{10}(f [\text{MHz}]) + 32.45$

5. Terrain Elevation Data & Processing

SRTM (Shuttle Radar Topography Mission)

Specifications:

- Resolution: 90 m (3 arc-seconds)
- Vertical accuracy: ± 16 m (90% confidence)
- Coverage: $\pm 60^\circ$ latitude (most continents, not polar regions)
- Source: NASA/USGS public domain

Why SRTM?

- Global coverage at reasonable resolution
- Free and unrestricted
- Well-validated against GPS surveys
- Adequate for RF diffraction modeling (terrain features >90 m matter most)

Elevation Profile Extraction & Processing

Step 1: Define great-circle path from TX to RX

- Great-circle is the shortest distance on Earth's surface, accounting for Earth's curvature (important for distances >50 km).

Step 2: Fetch SRTM elevation along the path

- Query SRTM tiles (typically 4 tiles per continental path), cache locally to avoid redundant API calls, fallback to OpenTopography API if SRTM unavailable.

Step 3: Resample using 200-point contour-enhanced algorithm

Why resampling?

- SRTM raw data at 90 m posting gives ~1,000+ points over a 100 km path
- Computational cost is $O(N^2)$ where N = profile points
- Need to preserve critical features (ridges, valleys) whilst reducing point count

Contour-enhanced algorithm:

- 1. Start with dense profile (e.g., 1,000 raw SRTM points)
- 2. Identify critical points: local maxima (ridge crests), local minima (valley floors), inflection points
- 3. Mark these as "must keep"
- 4. Iteratively remove non-critical points, preserving shape
- 5. Target: ~200 final points

Result:

- Maintains terrain shape and dominant obstacles
- Removes noise and small-scale roughness
- Balanced computational cost vs. diffraction accuracy

Step 4: Pass to propagation engine

- MF/LF: P.368-9 calculates diffraction from 200-point profile
- VHF/UHF: Deygout method identifies dominant obstacle from 200 points

Elevation Data Fallback

If SRTM unavailable (gaps in data, remote region):

- 1. Try OpenTopography API (slower, remote call)
- 2. If OpenTopography also unavailable: Use flat terrain (h = sea level)

Flat terrain assumption degrades accuracy significantly (± 5 – 10 dB error possible in hilly regions). Always check SRTM coverage before planning in remote areas.

6. Frequency & Validation Range

Input Acceptance Range

Minimum: 0.01 MHz (10 kHz)

Maximum: 30,000 MHz (30 GHz)

The app will compute propagation at any frequency in this range.

Recommended & Validated Range: 300 kHz – 2 GHz

Below 300 kHz (Extended LF/VLF):

- ITU-R P.368-9 model is still valid
- SRTM elevation coverage becomes sparse
- Suggested: Use only in regions with known elevation data
- Applications: LW broadcast (150–280 kHz), LF navigation beacons

300 kHz – 30 MHz (LF/MF/HF):

- Fully validated
- P.368-9 model + SRTM coverage complete
- Applications: AM broadcast, maritime, amateur radio, aviation HF

30 MHz – 2 GHz (VHF/UHF):

- Fully validated
- Fresnel-Kirchhoff diffraction + SRTM coverage
- Applications: FM broadcast, mobile cellular, TV, WiFi, satellite

2 GHz – 30 GHz (Microwave & Millimeter-wave):

- Model mathematically valid (Fresnel diffraction applies to any frequency)
- NOT validated against real-world measurements in this range
- Suggested: Use with caution; treat results as experimental
- Applications: 5G, satellite (Ku/Ka band), millimeter-wave (research only)

The 30 MHz Boundary

At 30 MHz, the app transitions from P.368-9 ground wave to Fresnel-Kirchhoff diffraction:

- < 30 MHz: Use P.368-9 Millington (ground wave + diffraction)
- $\geq$ 30 MHz: Use Fresnel-Kirchhoff (LOS + diffraction + two-ray)

This boundary reflects the physical transition from ground-wave-dominated to LOS-dominated propagation and aligns with ITU-R P.368-9's specified range (0.15–30 MHz).